REFONTE INTERNATIONAL TRAINING AND INTERNSHIP PROGRAM

Astrodynamics Specialist

SESSION 9

Re-entry Dynamics and Mission End-of-Life Planning

Coming Home: Deceleration, Heating, the Re-entry Corridor and Safe Disposal

Prepared for Cao Khanh Tan

Module 9, in the Advanced Application Phase

Builds on Session 2 (drag), Session 3 (delta-v), Session 6 (disposal)

Tools STK Lifetime and re-entry tools, GMAT, Python

SESSION 9

Re-entry Dynamics and Mission End-of-Life Planning

Everything so far has kept spacecraft in space. This module brings them down. Re-entry is the violent conversion of a spacecraft orbital energy into heat as it plunges into the atmosphere, and it is governed by a tight balance: too steep and the vehicle is crushed or burned; too shallow and it skips back into space. The same physics decides how a spent satellite is safely disposed of at the end of its life, now a legal requirement. This module covers the deceleration and heating of entry, the re-entry corridor, and end-of-life planning, tying together the drag of Session 2, the delta-v of Session 3, and the disposal of Session 6.

Learning objectives

- Define the ballistic coefficient and its role in how a body decelerates.
- Use the Allen-Eggers solution to find the velocity and deceleration through entry.
- Show that peak deceleration depends on entry angle, not on the ballistic coefficient.
- Explain aerodynamic heating and why re-entry vehicles are blunt.
- Describe the re-entry corridor and its undershoot and overshoot boundaries.
- Compute a deorbit burn and outline end-of-life disposal rules.

> **The two extremes**
> A ballistic capsule enters steeply: a short, brutal ride of high g and intense but brief heating. A lifting vehicle enters shallow: a long, gentle glide that spreads the heat over many minutes. Every re-entry design lives somewhere on that line between deceleration load and heat load.

PART 1

Deceleration: The Allen-Eggers Solution

How hard a body decelerates depends on how it trades mass against drag area, the ballistic coefficient (note the entry convention, mass over drag area, the inverse of the drag term in Session 2):

$$\beta = \frac{m}{C_D A}$$

A high ballistic coefficient (heavy, small, streamlined) punches deep into the atmosphere before slowing; a low one (light, broad, blunt) decelerates high up. With an exponential atmosphere and a constant flight-path angle, Allen and Eggers solved the entry in closed form. The velocity falls as:

$$\rho = \rho_0 \, e^{-h/H}$$

$$v(h) = v_e \exp\left[-\frac{\rho_0 H}{2\beta \sin\gamma} \, e^{-h/H} \right]$$

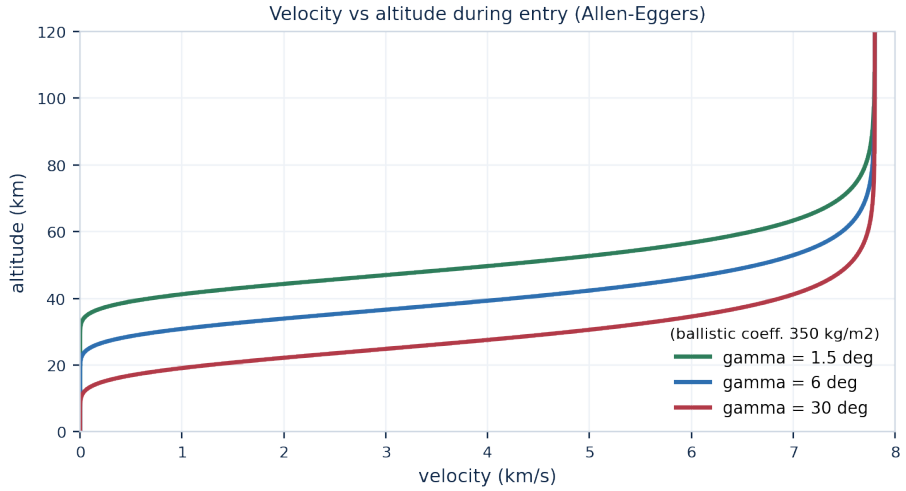


Figure 1. Velocity versus altitude for three entry angles at a fixed ballistic coefficient. A steep entry (30 deg) sheds its speed low and fast; a shallow one (1.5 deg) bleeds off gently over a long descent.

The deceleration is just the drag divided by mass, and its peak has a remarkable form:

$$a = \frac{\rho v^2}{2\beta}$$

$$a_{\max} = \frac{v_e^2 \sin\gamma}{2eH}$$

The Allen-Eggers surprise

Peak deceleration depends only on entry velocity, flight-path angle and the atmospheric scale height, not on the ballistic coefficient. A heavy capsule and a light one entering at the same angle feel the same peak g; the ballistic coefficient only sets the altitude at which it happens. At orbital speed, a 1.5 deg entry peaks near 4 g, a 6 deg entry near 16 g, and a 30 deg ballistic plunge near 76 g.

PART 2

Aerodynamic Heating

The energy that deceleration removes has to go somewhere, and most of it becomes heat in the shock layer ahead of the vehicle. The stagnation-point heating rate rises with the cube of velocity and the square root of density, and falls with the nose radius:

$$\dot{q} = k\sqrt{\frac{\rho}{R_n}}\, v^3$$

Two consequences shape every entry vehicle. First, because heating scales with the inverse square root of nose radius, a blunt body heats far less than a sharp one, which is why capsules are rounded, not pointed: the blunt shock stands off and pushes the hottest gas away from the surface. Second, peak heating and peak deceleration do not coincide, as Figure 2 shows.

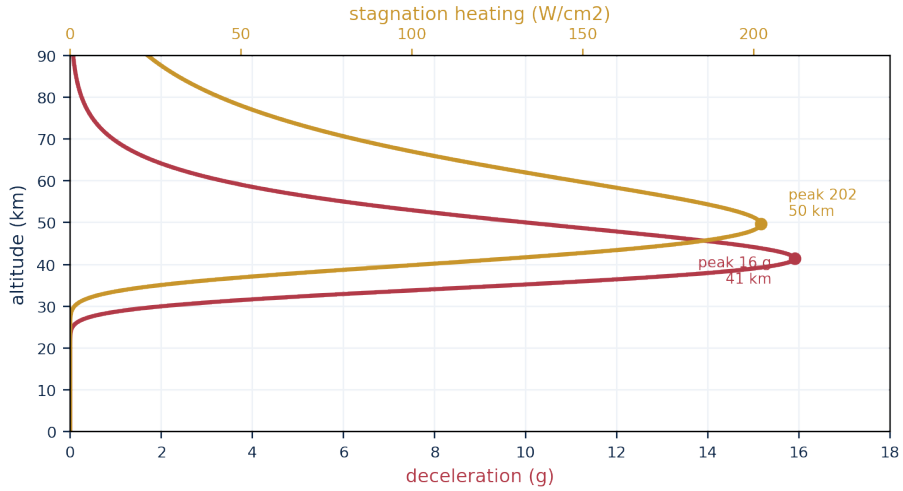


Figure 2. Deceleration (red) and stagnation heating (gold) through a capsule entry. Peak heating comes higher and earlier (near 50 km) than peak deceleration (near 41 km), because heating depends on velocity cubed and the vehicle is still fast when it first meets significant air.

Rate versus load

A steep entry gives a high peak heating rate but a short exposure, so a small total heat load. A shallow entry gives a lower peak rate but a long exposure, so a large total load. Thermal protection must survive both the peak rate and the integrated load, which pull the design in opposite directions.

PART 3

The Re-entry Corridor

The entry angle must fall within a narrow band, the re-entry corridor, bounded on both sides:

- **Undershoot boundary (too steep):** the vehicle exceeds its deceleration or heating limits and is destroyed.
- **Overshoot boundary (too shallow):** the atmosphere cannot capture the vehicle; it skips back out into space, like a stone off water.



Figure 3. The re-entry corridor. A nominal entry (green) stays inside the safe band; too steep (red) plunges into excessive g and heating; too shallow (gold) dips into the atmosphere and skips back to space.

Aerodynamic lift widens the corridor. A vehicle with a usable lift-to-drag ratio can adjust its trajectory during entry, pulling up if too steep or pushing down if too shallow, which is how

crewed vehicles keep deceleration tolerable and land accurately. A purely ballistic capsule has a much narrower corridor and must hit its entry angle precisely.

PART 4

End-of-Life Planning and Disposal

The same descent physics governs how a spacecraft is retired. Leaving dead satellites in busy orbits is now prohibited, and every mission must plan its disposal:

- **Low orbits: deorbit.** A retrograde burn lowers the perigee into the atmosphere so drag completes the re-entry. Large vehicles are targeted for a controlled re-entry over open ocean (the South Pacific "Point Nemo"); small ones may decay naturally. Guidelines now require clearance within about five years of end of mission.
- **Geostationary: graveyard.** There is no atmosphere to deorbit into, so the satellite is raised a few hundred kilometres above the ring, as in Session 6.
- **Design for demise:** spacecraft are increasingly designed so components burn up completely on re-entry, keeping the ground-casualty risk below accepted limits.

The deorbit burn is a small retrograde impulse that drops the perigee to a re-entry altitude:

$$\Delta v_{\text{deorbit}} = v_{\text{circ}} - v_a$$

> **Worked example: a deorbit burn**
> To bring a satellite down from a 400 km circular orbit, a single retrograde burn that lowers the perigee to about 50 km costs roughly 100 m/s. From 800 km it costs about 210 m/s. This is why low missions reserve a modest propellant margin for disposal, and why leaving it too late, once the propellant is gone, turns a satellite into long-lived debris.

Computing Entry and Deorbit

The Allen-Eggers profile and the deorbit burn are both short calculations; the entry curves in this module came from the code below.

```
import numpy as np
ve, H, rho0 = 7800.0, 7500.0, 1.225        # entry speed, scale height, sea-level density
beta, gamma = 350.0, np.radians(6)          # ballistic coeff (m/CdA), flight-path angle

h = np.linspace(120e3, 0, 600)
rho = rho0*np.exp(-h/H)
v = ve*np.exp(-(rho0*H)/(2*beta*np.sin(gamma))*np.exp(-h/H))
a = rho*v**2/(2*beta)/9.81                    # deceleration in g
q = 1.7415e-4*np.sqrt(rho/1.0)*v**3/1e4      # stagnation heating W/cm^2
a_max = ve**2*np.sin(gamma)/(2*np.e*H)/9.81  # ~15.9 g, independent of beta

# deorbit: retro burn from 400 km circular to a 50 km perigee
mu, Re = 398600.4418, 6378.137
ra, rp = Re+400, Re+50; at = (ra+rp)/2
dv = (np.sqrt(mu/ra) - np.sqrt(mu*(2/ra - 1/at)))*1000   # ~102 m/s
```

Production tools
STK Lifetime estimates natural decay and re-entry time from a given orbit and ballistic coefficient, and its re-entry and deorbit sequences (with GMAT) model the descent with full atmosphere and aerodynamics. The Allen-Eggers hand result is the sanity check: if a tool predicts a peak-g wildly different from v_e squared sin gamma over 2 e H, question the setup.

PRACTICE

This Week (aim for 12 to 15 hours)

Bring your work and plots to the next live session. As always, show your reasoning, not just the final numbers.

Assignment: Re-entry Dynamics and End-of-Life Planning

- **Reading and notes (3 to 4 h).** Explain why peak deceleration is independent of the ballistic coefficient, and why re-entry vehicles are blunt rather than pointed. One page.
- **Peak deceleration (3 to 4 h).** For an orbital entry at 7.8 km/s, compute the peak deceleration (in g) for flight-path angles of 2, 5 and 10 degrees using the Allen-Eggers formula, and comment on which are survivable for crew.
- **Entry profile (3 to 4 h).** Using the code in the notes, plot velocity, deceleration and stagnation heating versus altitude for a capsule (ballistic coefficient 300, entry angle 6 degrees). Identify the altitudes of peak-g and peak heating.
- **Deorbit budget (2 h).** Compute the retrograde delta-v to deorbit from a 600 km circular orbit to a 50 km perigee. Compare it to the 400 km case.
- **STK (2 h).** Use the STK Lifetime or re-entry tool to estimate the decay time or entry conditions for a low orbit, and capture the result.

Answer key (for your self-check)

Quantity	Expected value
Peak g at 2 / 5 / 10 deg (v = 7.8 km/s)	about 5.3 / 13.3 / 26.4 g
Peak-g altitude (capsule, 6 deg)	about 40 to 42 km
Peak heating altitude (capsule, 6 deg)	about 50 km (higher than peak-g)
Deorbit from 600 km to 50 km perigee	about 155 m/s (vs about 102 m/s from 400 km)
Why blunt	heating scales as 1/sqrt(nose radius): blunt = far less heating

REFERENCE

One-Page Formula Sheet

Ballistic coefficient (entry)	$\beta = \frac{m}{C_D A}$
Exponential atmosphere	$\rho = \rho_0 \, e^{-h/H}$
Allen-Eggers velocity	$v(h) = v_e \exp\left[-\frac{\rho_0 H}{2\beta \sin\gamma} \, e^{-h/H}\right]$
Deceleration (drag / mass)	$a = \frac{\rho v^2}{2\beta}$
Peak deceleration	$a_{\max} = \frac{v_e^2 \sin\gamma}{2\,e\,H}$
Stagnation heating	$\dot{q} = k\sqrt{\frac{\rho}{R_n}}\, v^3$
Deorbit delta-v	$\Delta v_{\text{deorbit}} = v_{\text{circ}} - v_a$

Key glossary

- **Ballistic coefficient:** mass over drag area; high values penetrate deeper before slowing.
- **Allen-Eggers solution:** the closed-form entry for a constant flight-path angle in an exponential atmosphere.
- **Re-entry corridor:** the band of entry angles between undershoot (too steep) and overshoot (skip-out).
- **Design for demise:** building spacecraft so they burn up fully on re-entry, limiting ground risk.
- **Point Nemo:** the remote South Pacific ocean area targeted for controlled re-entries.

Going further

For depth, see Vallado (re-entry and decay), Regan and Anandakrishnan, Dynamics of Atmospheric Re-Entry, and the classic Allen and Eggers NACA report. These are real, widely used references; please confirm editions yourself, as I cannot verify those details from here.

Coming up in Session 10
Optimization methods for mission design: how to find the best trajectory or constellation, not just a working one. Cost functions, constraints, and the numerical methods that search a design space, the final taught module before the capstone.